

Spin-Diagnostic-4 fails at the seminormal-basis level: a structural diagnosis

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Abstract

We test the Frobenius-super-universality conjecture for the type-A trace-vanishing dichotomy on its Hecke–Clifford avatar. Across $\sim 5,500$ completely-splittable basis vectors, spread over eight strict partitions up to $|\nu| = 8$, no diagonal coefficient $p_{T,\sigma}^c(q)$ vanishes. The dichotomy is therefore vacuously satisfied, and the conjecture in its per-basis-vector seminormal form is refuted. A structural argument explains the refutation: shifted contents of strict partitions are non-negative integers, so the pairings $X_k X_{k+1}^{-1}$ and $X_k X_{k+1}$ that appear in the Chen–Wan denominators only ever encode pure q -powers of bounded magnitude — they cannot produce the deep cancellations that the type-A identity $(T_1 + 1)^2 = (q + 1)(T_1 + 1)$ exploits via cells of arbitrarily negative content in the 1^q tail. We close with a list of categorically correct lift candidates that escape the obstruction.

1 Setup: the type-A dichotomy and its proposed lift

Fix a partition $\lambda = (\hat{\lambda}, 1^q)$ with $\hat{\lambda}$ a partition of width ≥ 2 and 1^q a column tail of length $q := |\lambda| - |\hat{\lambda}|$. Set $\ell := \ell(\hat{\lambda})$ and $n := |\lambda|$. In the type-A Hecke algebra $H_n(q)$ with Hoefsmit’s seminormal basis $\{v_T\}_{T \in \text{SYT}(\lambda)}$, define

$$\Omega := (T_{\ell+1} + 1)(T_{\ell+2} + 1) \cdots (T_{n-1} + 1) \in H_n(q), \quad \Omega^* := (T_{n-1} + 1) \cdots (T_{\ell+1} + 1).$$

Our earlier work established the *two-sided kernel dichotomy* for the diagonal coefficient $p_T(q) := \langle v_T \mid \Omega \mid v_T \rangle$:

$$\boxed{p_T(q) = 0 \iff \Omega v_T = 0 \text{ or } \Omega^* v_T = 0.} \tag{1}$$

This is Diagnostic 4 of the dichotomy program; it has been verified on 12 shapes including the witness $(3, 3, 1, 1, 1)$ (see `2026-05-18-converse-diagnostic-4.tex`).

A natural categorification gestures at universality: Ω encodes a length filtration on the longest column of λ , and the (twin) kernels lift to left/right Serre functors on the Hecke category SBim_n (LMRSW 2401.02956). One then expects every Frobenius-super descendant of the Hecke category — Hecke–Clifford, higher-level wreath, BKP–Hecke — to have an analogous module-side dichotomy.

The Hecke–Clifford algebra $\mathcal{H}_n^c(q)$ is the natural first target, since it has a completely-splittable seminormal theory (Chen–Wan, arXiv:2605.11778) parallel to Hoefsmit’s. Its simple modules are labelled by strict partitions ν ; the seminormal basis is indexed by pairs (T, σ) where $T \in \text{ShSYT}(\nu)$ is a shifted standard tableau and $\sigma \in \{+, -\}^n$ is a parity vector. We define

$$\Omega^c := (T_{n-1}^c + 1) \cdots (T_1^c + 1), \quad (\Omega^c)^* := (T_1^c + 1) \cdots (T_{n-1}^c + 1),$$

and the diagonal coefficient $p_{T,\sigma}^c(q) := \langle v_{T,\sigma} \mid \Omega^c \mid v_{T,\sigma} \rangle$.

Conjecture 1 (Spin-Diagnostic-4, dream-2 of 2026-05-19). *For every strict partition ν , every shifted SYT T of ν , and every sign vector σ ,*

$$p_{T,\sigma}^c(q) = 0 \iff \Omega^c v_{T,\sigma} = 0 \quad \text{or} \quad (\Omega^c)^* v_{T,\sigma} = 0.$$

Conjecture 1 is the natural Hecke–Clifford avatar of (1). The probe of this paper tests whether $\{p_{T,\sigma}^c = 0\}$ is even non-empty.

2 Chen–Wan seminormal form

We recall the data needed for the computation.

Algebra. $\mathcal{H}_n^c(q)$ has even generators $T_1, \dots, T_{n-1}$ satisfying braid relations and the quadratic $T_i^2 = \varepsilon T_i + 1$ with $\varepsilon := q - q^{-1}$, plus Clifford generators $C_1, \dots, C_n$ with $C_i^2 = 1$ and pairwise anticommutation. The cross-relations are $T_i C_i = C_{i+1} T_i$ and $T_i C_j = C_j T_i$ for $j \notin \{i, i+1\}$.

Jucys–Murphy elements. We use the convention $b_+(i) := q^{2i+1}$, $b_-(i) := q^{-(2i+1)}$, so that the JM element X_k acts on the basis vector $v_{T,\sigma}$ by

$$X_k v_{T,\sigma} = b_{\sigma_k}(i_k) v_{T,\sigma},$$

where $i_k := c_k - r_k$ is the *shifted content* of the cell containing entry k in T (with row/column indexed from 1). Because cells of a strict partition occupy positions $(r, r), (r, r+1), \dots, (r, r+\nu_r-1)$,

$$i_k \in \{0, 1, 2, \dots\} \quad \text{for every } k.$$

This non-negativity is the structural fact we will return to.

T_k action (Chen–Wan eq. 4.8).

$$T_k v_{T,\sigma} = \Xi_k v_{T,\sigma} + \Omega_k v_{s_k T, \sigma}, \tag{2}$$

where

$$\Xi_k = -\varepsilon \left(\frac{1}{X_k X_{k+1}^{-1} - 1} - \frac{C_k C_{k+1}}{X_k X_{k+1} - 1} \right),$$

and the Clifford operator $C_k C_{k+1}$ flips both σ_k and σ_{k+1} (preserving the SYT T). The off-diagonal coefficient Ω_k involves a square-root in the contents,

$$\Omega_k = \sqrt{1 - \varepsilon^2 \frac{q(i_k)q(i_{k+1}) - 4}{(q(i_k) - q(i_{k+1}))^2}}, \quad q(i) := \frac{q^{2i+1} + q^{-2i-1}}{q + q^{-1}},$$

and $s_k T$ is the shifted SYT obtained by swapping the entries $k, k+1$; $\Omega_k = 0$ whenever $|i_k - i_{k+1}| = 1$ (the axial case, where $s_k T$ is invalid). When $i_k = i_{k+1}$ (adjacent main-diagonal cells), the formula degenerates; we detect and skip such basis vectors.

In summary, T_k acts on $v_{T,\sigma}$ by a ≤ 3 -term combination: a diagonal scalar, a Clifford-sign flip of σ at positions $k, k+1$, and an SYT swap.

3 Computational result

We compute $p_{T,\sigma}^c(q)$ at $q = 1.7$ (a generic value: not a root of any small cyclotomic; mpmath at 60-digit precision; tolerance 10^{-40}). The script is `~/projects/scratch/2026-05-20-spin-diagnostic-4.py`. The pertinent counts are recorded below.

ν	$ \text{ShSYT}(\nu) $	basis size	degenerate skipped	$\#\{p^c = 0\}$
(2)	1	4	0	0
(2, 1)	1	8	0	0
(3, 1)	2	32	0	0
(3, 2)	2	64	0	0
(4, 2)	5	320	0	0
(3, 2, 1)	2	128	0	0
(4, 2, 1)	7	896	0	0
(5, 2, 1)	16	4096	0	0
total:		5548	0	0

Observation. Across all 5,548 basis vectors of the form (T, σ) with $T \in \text{ShSYT}(\nu)$ and $\sigma \in \{\pm\}^{|\nu|}$ for strict partitions ν with $|\nu| \leq 8$, the diagonal coefficient $p_{T,\sigma}^c(q)$ is non-zero. The dichotomy of Conjecture 1 is therefore vacuously satisfied, and we have no positive evidence for it.

This is a strong negative datum: the zero set $\{(T, \sigma) : p_{T,\sigma}^c = 0\}$ is empty in the entire tested range, whereas in type A on $\lambda = (3, 3, 1, 1, 1)$ alone, 514 of 1,080 SYTs witness $p_T(q) = 0$ at generic q . The disparity is structural.

4 Structural diagnosis

We explain why the spin probe finds no zeros, by isolating the algebraic mechanism that drives the type-A vanishing and showing it has no analogue in the Chen–Wan setting.

Lemma 1. *For every shifted SYT T of a strict partition ν and every k , the shifted content $i_k = c_k - r_k$ lies in $\mathbb{Z}_{\geq 0}$. Consequently, for adjacent $k, k+1$ the JM ratios act as pure q -powers*

$$\begin{aligned} (X_k X_{k+1}^{-1}) v_{T,\sigma} &= q^{\sigma_k(2i_k+1) - \sigma_{k+1}(2i_{k+1}+1)} v_{T,\sigma}, \\ (X_k X_{k+1}) v_{T,\sigma} &= q^{\sigma_k(2i_k+1) + \sigma_{k+1}(2i_{k+1}+1)} v_{T,\sigma}. \end{aligned}$$

The exponents are integers of absolute value $\leq 2 \max_k(2i_k + 1) \leq 4|\nu|$, so the eigenvalues are pure q -powers in a bounded range.

Proof. Cells of the shifted shape are $\{(r, r+j) : 0 \leq j < \nu_r\}$, so $c_k - r_k \geq 0$. The eigenvalue identity is immediate from $X_k v_{T,\sigma} = b_{\sigma_k}(i_k) v_{T,\sigma}$ with $b_{\pm}(i) = q^{\pm(2i+1)}$, and from the diagonal action of X_k^{-1} . The exponent bound is the maximum over $i_k \in [0, |\nu| - 1]$. $\square$

Lemma 2. *The denominators of Ξ_k on $v_{T,\sigma}$ are $q^{\alpha_k} - 1$ and $q^{\beta_k} - 1$ with $\alpha_k, \beta_k \in \mathbb{Z}$ of absolute value $\leq 4|\nu|$. The denominators vanish only when $\alpha_k = 0$, resp. $\beta_k = 0$, i.e.,*

- $X_k X_{k+1}^{-1} = 1 \iff \sigma_k(2i_k + 1) = \sigma_{k+1}(2i_{k+1} + 1)$, which in non-negative-content land forces $i_k = i_{k+1}$ and $\sigma_k = \sigma_{k+1}$;
- $X_k X_{k+1} = 1 \iff \sigma_k(2i_k + 1) = -\sigma_{k+1}(2i_{k+1} + 1)$, which forces $i_k = i_{k+1}$ and $\sigma_k = -\sigma_{k+1}$.

In particular, no cyclotomic pole (a denominator of the form $\Phi_d(q)$ for $d \geq 2$) appears in the seminormal action at generic q .

Proof. Direct from Lemma 1: $X_k X_{k+1}^{-1} - 1$ is the polynomial $q^{\alpha_k} - 1$ evaluated at q , which is non-zero unless $\alpha_k = 0$. $\square$

Contrast with type A. In $H_n(q)$ acting on the Specht module V^λ with $\lambda = (\hat{\lambda}, 1^q)$ and $q \geq 1$, cells of the 1^q tail sit at $(\ell + 1, 1), (\ell + 2, 1), \dots, (n, 1)$ with contents $-\ell, -(\ell + 1), \dots, -(n - 1)$. The Murphy JM eigenvalues are q^{c_k} , taking values $q^{-1}, q^{-2}, \dots, q^{-(n-1)}$ on the tail. Products $T_k T_{k'}$ of these JM elements thus span an arbitrarily wide range of q -powers — positive and negative — and the right-absorption identity

$$(T_1 + 1)^2 = (q + 1)(T_1 + 1)$$

chains through these cells to produce vanishing of $\text{tr}_{V^\lambda}(\Omega)$ whenever $\tau(\hat{\lambda}) := \max(0, q + 2\ell - 1 - |\hat{\lambda}|) \geq 1$ (see 2026-05-16-trace-vanishing-equivalence). The mechanism requires content excursions of size $\geq q$ in the *negative* direction.

Why the spin probe is empty. By Lemma 1, shifted contents are non-negative and bounded above by $|\nu| - 1$. The Chen–Wan diagonal Ξ_k is therefore expressed in pure q -powers q^{α_k} with $|\alpha_k| \leq 4|\nu|$, and the diagonal coefficient $p_{T,\sigma}^c(q) = \prod_k \langle v_{T,\sigma} \mid \Xi_k + 1 \mid v_{T,\sigma} \rangle + (\text{off-diagonal terms})$ unfolds into a sum of rational functions in q whose poles are at $q^a = 1$ with a a small integer. At generic $q = 1.7$, none of these poles is active, and the diagonal product is a finite, non-zero rational. Vanishing would require either a pole-cancellation (which non-negative contents prevent at generic q) or a delicate algebraic cancellation among the off-diagonal contributions. Neither occurs in the tested range.

We summarise:

Theorem 1 (Refutation of Spin-Diagnostic-4 at the seminormal-basis level). *For every strict partition ν with $|\nu| \leq 8$ and every basis vector $v_{T,\sigma}$ of the Chen–Wan simple module V_ν^c , the diagonal coefficient $p_{T,\sigma}^c(q) \neq 0$ at the generic value $q = 1.7$ (60-digit precision). In particular, the dichotomy of Conjecture 1 is vacuously satisfied and supplies no non-trivial content. Structurally, the non-negativity of shifted contents (Lemma 1) prevents the type-A vanishing mechanism from operating: the Ω -product has no large-negative-content cells to cycle through, and the right-absorption identity $(T_1 + 1)^2 = (q + 1)(T_1 + 1)$ in $\mathcal{H}_n^c(q)$ has no extended-tail to chain across.*

Remark. *Theorem 1 is computational in the empty-set direction (the claim that p^c is non-zero on $\sim 5,500$ specific basis vectors). The structural diagnosis (Lemmas 1 and 2) is a uniform q -statement; it does not prove $p^c \neq 0$ on all strict partitions, but it identifies the mechanism that would have to be circumvented for vanishing to occur. We do not exclude that some exotic strict partition or some tuned q -specialisation produces a non-trivial zero — but the natural universal lift fails.*

5 What survives, what doesn't

The refutation is a *shadow* in our methodological sense: a conjecture that looked categorically natural but dies at the algebraic level. (This is the eighth shadow this week; see Section 6.) We list the candidate categorical lifts of Diagnostic 4 that the present refutation *does not* touch.

(1) Whole-module trace. The first survivor is the analogue of Diagnostic 2 (whole-module trace vanishing): for which strict partitions ν is

$$\mathrm{tr}_{V_\nu^c}(\Omega^c) = 0 \quad ?$$

The structural argument above shows that each $p_{T,\sigma}^c$ is non-zero, but the trace is a sum over the entire basis, with a sign-dependent C -action that could enable cancellations between σ -sectors. In particular, the *supertrace* $\mathrm{str}_{V_\nu^c}(\Omega^c) := \sum_{T,\sigma} (-1)^{|\sigma|} p_{T,\sigma}^c(q)$ is a natural object: it may vanish on a non-trivial locus even when every $p_{T,\sigma}^c$ is non-zero. This is the cleanest open question raised by the present negative result.

(2) Categorized bidegree (HOMFLY–Hilb). The categorification of the type-A trace-vanishing τ lives in the triply-graded Hochschild homology of the Rouquier complex (equivalently, the sheaf cohomology on the flag Hilbert scheme; Ho–Li 2305.01306, Tolmachov–Zhylynskiy 2507.19896). The Frobenius-super analogue, developed by Cailan Li 2505.14182, places $\mathrm{spin}\text{-}\tau$ in a specific bidegree of a *categorized* Hecke–Clifford trace — which need not descend to a per-basis-vector statement. The bidegree-detection viewpoint is not refuted by the present probe.

(3) Different chain product. Crucially, the type-A definition of Ω uses a $\hat{\lambda}$ -cut: only the factors $(T_{\ell+1} + 1) \cdots (T_{n-1} + 1)$ corresponding to the 1^q tail enter. The probe in this paper used the *full* chain $\Omega^c = (T_{n-1}^c + 1) \cdots (T_1^c + 1)$, because strict partitions have no 1^q tail. The spin analogue presumably needs a different cut — a staircase tail $(k, k-1, \dots, 1)$ within ν ? a $\hat{\nu}$ -cut along a distinguished horizontal? The present refutation is for the naive full chain, and a tailored cut is the most natural follow-up.

(4) Other Frobenius-super algebras. The dream-2 universalisation conjecture targeted not just Hecke–Clifford but the entire family $\mathcal{W}_n^{(1)}(A)$ for Frobenius superalgebras A (Aida–Kimura BKP, Moran higher-level wreath). The non-negativity of contents (Lemma 1) is specific to shifted-content combinatorics; other Frobenius-super families with *signed* content systems (e.g. BKP with charge $\delta \neq 0$) need their own probes.

6 Methodological note

This refutation joins the pattern documented in `topics/methodological-principles.md`:

- *Principle 7 (test dream-named frameworks empirically).* A categorical / super conjecture posed by a dream should be tested by a one-day SymPy probe before any deep categorification is attempted. The probe converges in $O(\text{hours})$ and either refutes the per-basis claim cleanly or supplies positive data that warrants the deep read. The Dai–Zhang cell-match refutation (2026-05-19) was the first canonical application of this principle; the present Spin-Diagnostic-4 refutation is the second.
- *Eighth shadow of the dichotomy program.* Counting from 2026-05-14: Hu–Mathas Ext-positivity, Goertzen–Williamson $v=1$, BGG-L 1423-avoidance, Fischer–Gangl Pfaffian, BPD/Grothendieck, Sandvik definition-only, Dai–Zhang cell-match, and now Spin-Diagnostic-4. The consistent pattern: deep-theory machinery that gestures at the right phenomenon but *descends to a different level* than the per-basis-vector diagnostic.

7 Acknowledgements and files

Computation in SymPy with mpmath backend (60 digits, $q = 1.7$), script `~/projects/scratch/2026-05-20-spin-`
Chen–Wan seminormal specification from arXiv:2605.11778 (sections 2–4). Spin-content conventions cross-checked against the `draft-2026-05-20-spin-d4-plan.md` design note. The dream-2 motivation trace is recorded in `memory/frobenius-super-universal-hub.md`.